Year 13 Further Maths - Non-parametric tests

Similar questions - Rank-sum and signed-rank tests

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A Wilcoxon rank-sum test is to be carried out with samples of sizes $m = 4$ and $n = 6$. The test statistic R_m is the sum of the ranks of the sample of size 4. Each possible set of 4 different integers from $1, 2, \ldots, 10$ is considered, and its sum S is found. Part of the frequency distribution is shown (out of 210 sets).

S	10	11	12	13	14	15	16
Frequency	1	1	2	3	5	6	9
Cumulative	1	2	4	7	12	18	27

(a) Use the table to find the critical region for a 1-tail test at the 2% significance level (small values of R_m indicate the effect), and the critical region at the 5% level. **[4]**

(b) A student says that for large m and n the mean and variance of R_m are 180 and 580 respectively. Show that at least one of these values must be incorrect. **[3]**

2 A college course was taught by two lecturers. Students chose which lecturer to attend. At the end of the course all students sat the same examination. A random sample of 25 students taught by Lecturer Q and 30 taught by Lecturer R had their marks ranked together. The sum of the ranks of the Q students was 600. Test at the 5% significance level whether there is a difference in the ranks of the two groups of students. **[9]**

3 Jo can use either route A or route B to school and believes that A has the shorter journey times. She ranks 15 journey times: 8 by route A and 7 by route B, in increasing order of time. The sum of the ranks of the journeys by route B is 75.

(a) Test at the 5% significance level whether route A has shorter journey times than route B. (You may use the fact that, for sets of 7 different integers chosen from 1 to 15, the sums W satisfy $\mathrm{P}(W \geqslant 71) = 0.0469$ and $\mathrm{P}(W \geqslant 70) = 0.0603$.) **[8]**

(b) State an assumption about the journeys that is necessary for the test to be valid. **[1]**

4 Nine pupils rated school lunches on a scale from 0 to 20 before and after a new menu was introduced.

Pupil	1	2	3	4	5	6	7	8	9
Before	8	12	5	14	9	11	7	13	10
After	11	10	9	19	14	10	12	17	16

(a) Carry out a Wilcoxon signed-rank test at the 5% significance level (two-tail) to determine whether pupils' opinions have changed. You may use $\mathrm{P}(W \leqslant 5) = 0.0195$ and $\mathrm{P}(W \leqslant 6) = 0.0273$ for 9 pairs. **[8]**

(b) State what must be true of the distribution of differences for this test to be valid. **[1]**

(c) Explain why a pairs test is more appropriate than a test using ranks of two independent samples. **[1]**

5 The critical region for an r% two-tailed Wilcoxon signed-rank test, based on a large sample of size n, is $\{W_+ \leqslant 81\} \cup \{W_+ \geqslant 219\}$.

(a) Show that $n = 24$. **[3]**

(b) Using a suitable approximation, determine the value of r. **[4]**

6 A historian believes that the average age at which men married in a certain century was higher in urban areas than in rural areas. A random sample of 8 urban and 6 rural men gave the following ages (in years, to 2 d.p.).

Urban	18.25	19.75	20.58	25.50	34.50	41.67	46.25	23.50
Rural	18.00	18.08	18.33	19.50	22.17	18.92		

(a) Use an appropriate non-parametric test at the 5% significance level (you may use $\mathrm{P}(R \leqslant 31) = 0.0406$ and $\mathrm{P}(R \leqslant 32) = 0.0539$, where R is the sum of the ranks for the smaller sample). **[7]**

(b) State one assumption needed for the test to be valid. **[1]**

Question 1

(a) $P(R_m \leqslant 12) = \frac{4}{210} = 0.019 \leqslant 0.02$ and $P(R_m \leqslant 13) = \frac{7}{210} = 0.033 > 0.02$, so the 2% critical region is $R_m \leqslant 12$. For 5%: $P(R_m \leqslant 13) = 0.033 \leqslant 0.05$ but $P(R_m \leqslant 14) = \frac{12}{210} = 0.057 > 0.05$, so the critical region is $R_m \leqslant 13$

(b) $E(R_m) = \dfrac{m(m+n+1)}{2} = 180 \Rightarrow m(m+n+1) = 360$. $\text{Var}(R_m) = \dfrac{mn(m+n+1)}{12} = 580 \Rightarrow mn(m+n+1) = 6960$.

Dividing: $n = \dfrac{6960}{360} = 19.33$, which is not an integer, so at least one value is incorrect

Question 2

H_0: no difference in the (population) ranks; H_1: there is a difference. With $m = 25$, $n = 30$: $E(R) = \dfrac{25(56)}{2} = 700$, $\text{Var}(R) = \dfrac{25 \times 30 \times 56}{12} = 3500$, sd $= 59.16$. $z = \dfrac{600 - 700}{59.16} = -1.69$. For a 2-tail 5% test the critical value is ± 1.96; -1.69 is not in the critical region, so do not reject H_0: no evidence of a difference

Question 3

(a) H_0: the populations of journey times are the same; H_1: route A times are shorter, so the B ranks are larger. 1-tail test: the sum of the ranks of B (smaller sample) $W = 75$. $P(W \geqslant 71) = 0.0469 \leqslant 0.05$ but $P(W \geqslant 70) = 0.0603 > 0.05$, so the critical region is $W \geqslant 71$. $75 \geqslant 71$, so reject H_0: there is evidence that route A has shorter journey times

(b) The journey times are independent (and the two distributions have the same shape / differ only by a shift)

Question 4

(a) H_0: median difference is 0 (no change); H_1: median difference is not 0. Differences (after − before): $3, -2, 4, 5, 5, -1, 5, 4, 6$. Ranks of $|d|$: $1 \to 1$, $2 \to 2$, $3 \to 3$, $4, 4 \to 4.5, 4.5$, $5, 5, 5 \to 7$ each, $6 \to 9$. $W_- = 1 + 2 = 3$, $W_+ = 42$. Test statistic $W = \min = 3$. For $n = 9$ the 2-tail 5% critical region is $W \leqslant 5$ ($P(W \leqslant 5) = 0.0195 < 0.025$, $P(W \leqslant 6) = 0.0273 > 0.025$). $3 \leqslant 5$, so reject H_0: evidence that opinions have changed (improved)

(b) The differences come from a symmetric distribution

(c) The data are naturally paired (same pupils), so the differences remove pupil-to-pupil variation

Question 5

(a) The critical region is symmetric about the mean $\dfrac{n(n+1)}{4} = \dfrac{81 + 219}{2} = 150$, so $n(n+1) = 600$ and $n = 24$ (since $24 \times 25 = 600$)

(b) $\text{Var} = \dfrac{n(n+1)(2n+1)}{24} = \dfrac{24 \times 25 \times 49}{24} = 1225$, sd $= 35$. $P(W_+ \leqslant 81) \approx P\left(Z \leqslant \dfrac{81.5 - 150}{35}\right) = P(Z \leqslant -1.957) = 0.0252$. Both tails: $r \approx 5.0$

Question 6

(a) Rank-sum test (Wilcoxon / Mann–Whitney). H_0: the populations of ages are the same; H_1: urban ages are higher, so the rural (smaller) sample has lower ranks. Ranks of all 14: rural ages receive $1, 2, 4, 5, 6, 9$, sum $R = 27$. Critical region: $R \leqslant 31$ ($P(R \leqslant 31) = 0.0406 \leqslant 0.05 < P(R \leqslant 32)$). $27 \leqslant 31$, so reject H_0: evidence that men married later in urban areas

(b) The two samples are independent random samples (and the distributions have the same shape)